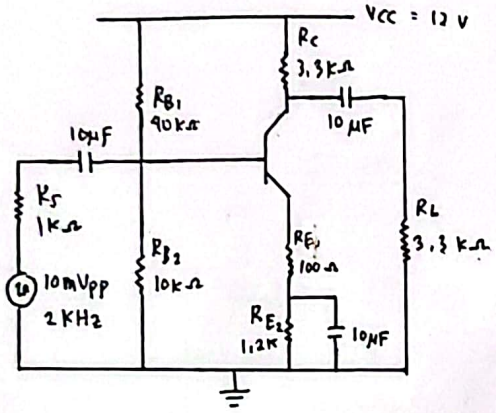
ELEKTRONIKA ANALOG

Nama : Devita Syifa Azzahraa

NIM : 221331009

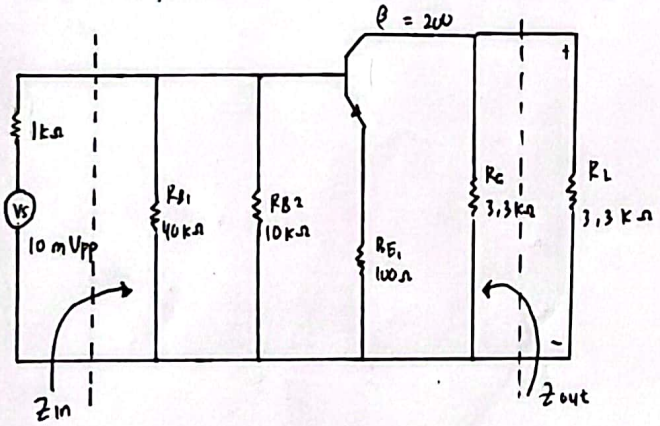
Kelas : 2A TC

Buat dan hitung rangkaian dibawah ini !

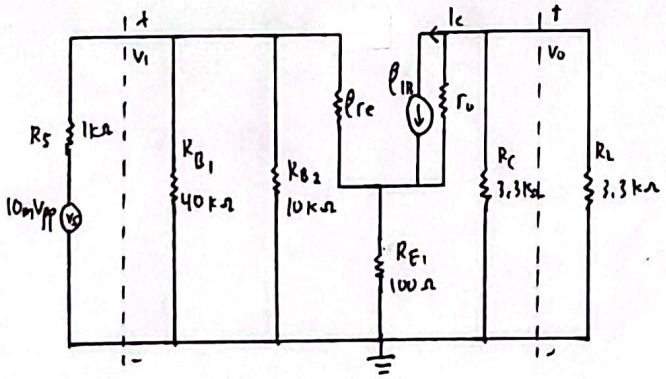


Jawab :

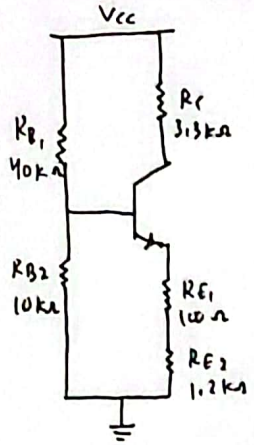
a. Buat rangkaian ekivalen ac (c = short; Vcc = 0V)



b. Ubah transistor dengan model re



C. Hitung dan gambar titik kerja dan garis beban DC



Jika $\beta re' > 10 R_{B2} \rightarrow (200)\ 1,3k > 10.10k$

$260k > 100k$

$\boxed{Ib \cong 0}$.

- $V_B = \frac{R_{B2}}{R_{B1} + R_{B2}} \cdot V_{CC} = \frac{10k}{40k+10k} \cdot 12 = 2,4\ V$
- $V_E = V_B - V_{BE} = 2,4 - 0,7 = 1,7\ V$

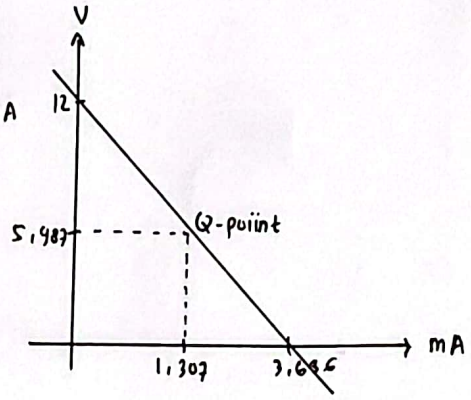
- $I_E = \frac{V_E}{R_E} = \frac{1,7}{1,3k} = 1,307\ mA$ (dengan $R_E = R_{E1} + R_{E2} = 100 + 1,2k\Omega = 1,3k\Omega$).
- $re = \frac{26\ mV}{I_E} = \frac{26\ mV}{1,307\ mA} = 19,89\ \Omega$
- $I_C = I_E = 1,307\ mA$
- $V_{CE} = V_{CC} - (I_C . R_C) - (I_E . R_E) = 12 - (1,307 . 3,3k) - (1,307 . 1,3k)$

  $= 12 - 4,3131 - 1,6991$

  $= 5,9878\ V$

Untuk garis beban, dibutuhkan :

- $I_{C\,satt} = \frac{V_{CC} - V_{CE}}{R_C} = \frac{12 - 0}{3,3k} = 3,636\ mA$
- $V_{CE}.\ Cut\ off = V_{CC} = 12\ V$



d. Hitung Av, Avs, Ai, Zi, Zo (lihat model re)

- $R_B' = R_{B1} // R_{B2}$

  $= \frac{40k . 10k}{40k + 10k} = 8k$

- $Z_i = R_{B1} // \beta re$

  $= 8k // (200)\ 19,89$

  $= 2,1656\ k\Omega$

- $A_V = \frac{V_o}{V_i} = \frac{-I_o . R_C}{Ib.\beta re + I_E \times R_E}$

  $= \frac{-R_C}{re + R_E'}$

  $= \frac{-3,3k}{19,89 + 100} = -27,52$

- $Z_o = R_C // r_o$ (dengan $r_o = \infty$)

  $= R_C = 3,3k\Omega$

- $A_V = \frac{I_o}{I_i} = \frac{I_C}{I_B} = \frac{\beta . Ib}{Ib} = 200$